

CIVE 202 – Project 5

Spring 2026

Scope of Work

Madalynne George

Gabriela Gonzalez Ng

Macy Falconer

Project Summary

Risk Averse, LLC has requested an analysis of one specific natural hazard across six different states. Selected locations are to be located close to one another to enable regional conclusions. A summary of how local populations were impacted by such disasters will also be provided. NRI definitions will be used to define risk for the analysis process. This definition will be clearly documented with references and justifications for the calculations. Additionally, Risk Averse LLC has asked that the NRI Census Tract data is used for analysis. To achieve these requests, a series of tasks and goals were outlined to aid in the execution of the requested analysis.

Project Goals

The goal of this project is to analyze the data per the requests of the client and present the findings in a manner that is easy to interpret and draw conclusions from. To do so, a plot for key trends in NRI data and a plot indicating impacted populations will be produced. To supplement these plots, a series of discussion papers will be produced. These discussions will address how the hazard was chosen, whether any regional trends were observed, and how local communities were impacted.

Project Tasks

The client has tasked us to make a summary plot which shows how the data for our hazard differs across different states. The plot needs to highlight key trends in the NRI data and should be easy for the client to interpret. An analysis also needs to be made, which will indicate how the natural hazard risk relates to population. Lastly, a discussion needs to be made which highlights how the risk of the natural hazard differs across 6 states, whether any regional trends were observed, and how the hazard has impacted communities.

Project Deliverables

The deliverables and due dates for this project are summarized in the following table:

Deliverable	File included	Due to Client by
Raw Data files	SVI_2022_US_county	4-27-26
Python Code File	Project #5.ipynb	4-27-26

Annotated Code Document	CIVE 202 – Annotated Code Project #5.pdf	4-27-26
Written Report	CIVE 202 – Project #5 Report.pdf	4-27-26